Citation Cliques in Questionable Journals

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March 2026

Abstract

This study examines whether questionable journals, defined through subject-normalized Eigenfactor percentiles, systematically foster denser and more reciprocating patterns of author-to-author citation. Using Crossref records, we assign journals to broad subject areas, compute subject-specific Eigenfactor scores, propagate venue quality to works and authors, match authors in questionable versus high-influence venues by subject and h_5, and analyze citation edges for cohesion and anomalies. Across 9,431 matched pairs, authors in questionable venues exhibit significantly higher cohesion: 6.7× higher co-author citation rates and 4.7× higher reciprocity in the aggregate Case–Control comparison. A subject-aware hybrid detection pipeline flags 277 outliers with 93.5% Case purity; these outliers display an 11× clique-strength lift relative to non-outliers, revealing a stark "Two Worlds" segregation ($r = 0.71$) where questionable venues operate as closed citation economies. The largest detected component ($n = 23$)

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displays a hub-and-spoke topology in which peripheral "Sycophants" funnel citations to central "Beneficiaries" through coordinated bursts, confirming a directed flow imbalance rather than reciprocal exchange among equals. Overall, cohesion, rather than broad asymmetry, accounts for the main case-control differences, suggesting that questionable venues foster segregated, inward-looking citation economies that distort bibliometric indicators.

Keywords: scientometrics; low-influence journals; citations; cliques; social network analysis; eigenfactor

1 Introduction

Quantitative indicators such as citation counts, the h-index, and journal metrics have become central to the assessment of scholarly impact. These measures offer convenient proxies for influence, but they also create incentives for strategic behavior and manipulation (Biagioli & Lippman, 2020). When deployed uncritically, citation-based indicators risk rewarding volume or visibility over substance, with unintended consequences for research integrity and evaluation systems.

Beyond well-known practices such as excessive self-citation, more coordinated behaviors have emerged. Authors may form tightly connected *citation cliques* –groups that disproportionately reference one another, often within low-influence journals as defined by subject-normalized Eigenfactor percentiles. Such practices distort indicators, undermine trust in the scholarly record, and complicate fair comparison across fields. These dynamics exemplify what has recently been described as *citation orchestration* (Evdaimon, Ioannidis, et al., 2024), where citations serve strategic rather than intellectual functions.

Questionable journals provide fertile ground for these dynamics. Their emphasis on publication volume, rapid turnaround, and relaxed editorial oversight creates environments where abnormal citation patterns can thrive. Systemic distortions such as national *home bias*

(Qiu et al., 2025) further interact with venue incentives, amplifying the risk that rankings and impact assessments reflect coordination rather than genuine uptake of ideas.

This study investigates whether questionable journals systematically foster cliques and abnormal citation patterns. Using large-scale bibliographic data, we implement a subject-aware pipeline that links journal influence to authors, matches Bottom- and Top-tier researchers of comparable productivity (by h_5 index), and analyzes their citation networks for cohesion and anomalies.

We find that extreme anomalies cluster exclusively among Bottom-tier authors in questionable venues: they concentrate in unusually dense, inward-looking structures and form compact outlier groups that set them apart from their Top-tier counterparts. These results provide empirical evidence that venue incentives can shape citation behavior in ways that distort bibliometric indicators. Throughout this paper, authors publishing primarily in low-influence venues are referred to as *cases*, and their matched counterparts in high-influence venues as *controls*. Terms such as "syndicate," "sycophant," and "beneficiary" are used as *structural labels* describing observed network roles and do not imply intent or misconduct.

2 Relevant Work

Prior research has highlighted the fragility of citation-based indicators when used as proxies for scholarly quality. Performance measures not only shape incentives but also invite gaming, from excessive self-citation to coordinated cartels (Biagioli & Lippman, 2020). The wider literature on the "metrics culture" documents how quantitative indicators reshape scholarly communication, creating feedback loops between evaluation systems and researcher behavior. These mechanisms underpin the emergence of citation cartels, inflated impact factors, and other forms of metric manipulation that erode the interpretive value of bibliometric indicators.

Venue-level frameworks were developed to mitigate some of these distortions. The Eigenfactor metric (Bergstrom et al., 2008) introduced a network-based approach that

discounts journal self-citations and weights each citation by the prestige of the citing venue. This recursive weighting sought to capture influence rather than popularity and to counteract local citation loops. However, later analyses have shown that such corrections remain sensitive to database coverage, field size, and reference density, indicating that structural biases persist even in sophisticated venue-level measures.

At the author level, organized citation exchanges have been formalized as *citation orchestration* (Evdaimon, Ioannidis, et al., 2024). This framework quantifies concentrated citation flows through indicators such as C/h^2, $A50\%C$, and $A50$, revealing unusually cohesive collaboration networks that may signal collective metric inflation. Earlier studies of citation cartels and excessive self-referencing align with this view, emphasizing how reciprocal ties among small groups of authors can yield disproportionate bibliometric visibility. Such coordinated behaviors blur the line between collaboration and manipulation, challenging the assumption that citation networks represent organic intellectual influence.

Systemic and infrastructural distortions further complicate these dynamics. National *home bias* inflates domestic citation counts, particularly in large scientific systems such as China's (Qiu et al., 2025), while technical manipulation has emerged through "sneaked references" inserted into metadata but absent from article text (Besançon, Cabanac, et al., 2024). Industrial-scale paper mills produce fabricated papers that populate citation networks with artificial linkages (Abalkina et al., 2025). Recent investigations reveal even more elaborate schemes, including the creation of fictitious author identities and peer reviewers to infiltrate legitimate journals. As reported by Naddaf (2025), these fake personas –sometimes maintained across multiple journals– enable mills to generate both papers and favorable reviews, prompting publishers to explore researcher identity verification frameworks and persistent identifier checks such as ORCID trust markers.

Finally, the recognition of complex citation distortions has increased the focus on transparency and reproducibility. The *LLM Guidelines* (LLM Guidelines Consortium, 2024) advocate for documenting decision rules and uncertainty, which is critical as machine learning

becomes central to integrity audits. Supporting initiatives promote open data and verifiable auditing to prevent large-scale manipulation. These developments suggest that citation systems are socio-technical infrastructures where credibility relies on both methodological openness and quantitative rigor.

The present research extends this discourse by systematically examining the relationship between author-level citation behaviors and the influence level of the publication venue, while controlling for individual productivity (h_5 index) and disciplinary norms through a matched case-control design. While previous studies have primarily investigated the orchestration of specific bibliometric indicators (Evdaimon, Ioannidis, et al., 2024) or semantic irregularities within article metadata (J. Liu et al., 2024), this study identifies *structural cohesion*, which manifests as dense local clustering and triadic closure, as the primary diagnostic signature of citation anomalies within low-influence journals. By implementing a subject-aware isolation forest on author-to-author networks across 9,431 matched pairs, the analysis demonstrates that such cliques constitute structural byproducts of venue-level incentives rather than isolated asymmetric flows. Consequently, this approach provides a network-centric diagnostic framework for identifying organized citation practices that evade traditional bibliometric oversight by focusing on network closure patterns rather than simple citation counts.

3 Methodology

Overview. We developed a subject-aware pipeline linking journal quality to author citation behavior. The main steps were: (i) journal subject classification, (ii) computation of subject-specific Eigenfactor scores, (iii) propagation of venue quality to works and authors, (iv) matched pairing of Bottom- and Top-tier authors by productivity, and (v) construction of author-author citation networks with cohesion and anomaly features. Figure 1 illustrates the main analysis subgraph of this pipeline.

To ensure consistency across all stages of the data pipeline, intermediate database layers

were validated through *Relational Database Unit* (RDBUnit) testing (Spinellis, 2024), which systematically checks referential integrity, record completeness, and join consistency before statistical analysis.

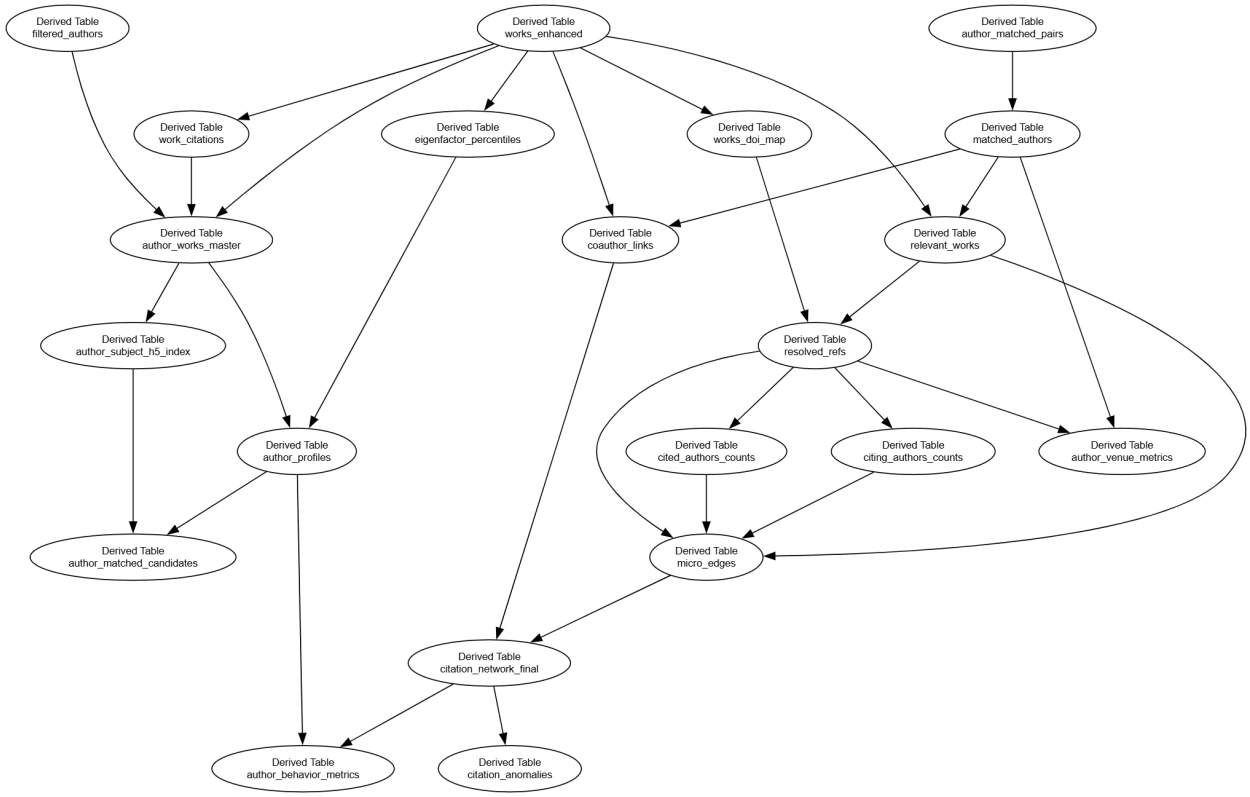

Figure 1: Analysis DAG for the social-network stage. Ellipses represent derived tables and arrows indicate dependencies. Curated inputs feed `citation_network_final` and downstream aggregates.

3.1 Data and Preprocessing

Data collection. Crossref metadata were ingested via *alexandria3k* (Spinellis, 2023), covering works published between 2020 and 2024. We retained DOIs, years, titles, abstracts, references, author identifiers (ORCID), and normalized ISSNs for journals. Duplicates and inconsistencies were cleaned, and derived layers linked works to authors, journals, and citations. Appendix C documents the full data derivation pipeline and the filtering decisions applied at each stage.

Journal Subject Classification. Each journal was assigned to one of five broad subjects: 1) Health Sciences, 2) Life Sciences, 3) Multidisciplinary, 4) Physical Sciences, 5) Social

Sciences & Humanities, using an LLM-based classifier (LLM Guidelines Consortium, 2024). We used `gpt-4.1` with zero temperature and few-shot exemplars, drawing on representative titles and abstracts. Prompts and decision rules prioritized abstracts over titles and journal names. To ensure reproducibility, we used seed and cached predictions to JSON files for safety.

Eigenfactor Computation. To quantify journal influence, we computed subject-specific Eigenfactor scores (Bergstrom et al., 2008) on journal citation networks (2020-2024), excluding self-citations. The computation used standard power iteration and produced normalized influence scores within each subject.

Defining Venue Quality. Within each subject, journals in the top quartile of Eigenfactor were labeled *high-influence*, and those in the bottom quartile *questionable*. Labels propagate to works (by ISSN) and to authors through their portfolios. Alternative thresholds and persistence checks were evaluated for robustness.

3.2 Author Matching and Network Construction

Author Tiering and Matching. Authors were classified as **Top** if most of their output was in high-influence journals, **Bottom** if concentrated in questionable journals, and **Mixed** otherwise. Mixed-tier authors were excluded from the matched-pair analysis to ensure clean separation between venue quality levels. We then matched Bottom authors to Top authors of similar productivity (subject-specific h_5 index), ensuring comparability in output volume and field.

Sample and Availability. The final analytical dataset comprises 9,431 matched case-control author pairs (see Table 6 for the per-subject breakdown). All pair-level and network-level aggregates were computed per subject area and combined only after validation of within-subject consistency. All processed data tables are available through the public replication repository described in Section 10.

Citation Network and Metrics. For matched authors, we constructed directed author-

to-author citation networks (2020–2024). Each citation edge between a citing work with n_c authors and a cited work with n_a authors received weight $w = 1/(n_c \times n_a)$, correcting for team size. From these edges we derived cohesion metrics (self-citation, co-author citation, clustering, triangles, k-core, PageRank) and pairwise anomaly metrics (reciprocity, asymmetry, velocity, citation bursts). These measures capture both dense reciprocal structures and asymmetric one-way flows. During exploratory analysis, raw pairwise metrics (maximum asymmetry, average velocity, and maximum burst) were also computed; in the formal pipeline they were replaced by normalised variants that are comparable across activity levels: *citation balance* captures net giver/receiver bias as a bounded $(-1, +1)$ ratio, while *normalised burst intensity* adjusts the largest single-peer citation weight by total incoming volume. Table 4 (Appendix A) lists the 14 features retained for statistical testing and modelling.

3.3 Statistical Analysis and Outlier Detection

Statistical Analysis and Robustness. Matched pairs were compared using Wilcoxon signed-rank tests (Wilcoxon, 1945), with Benjamini-Hochberg FDR correction (Benjamini & Hochberg, 1995). Effect sizes were quantified via Cliff's δ (Cliff, 1993) and Cohen's d (Cohen, 1988), with confidence intervals estimated through bootstrap resampling (Efron & Tibshirani, 1993). Robustness checks included permutation tests, placebo pairing, random-effects meta-analysis across subjects, and alternative outlier detectors (Isolation Forest (F. T. Liu et al., 2008), Local Outlier Factor, One-Class SVM). Sensitivity analysis utilized z-score thresholds $(1\sigma, 2\sigma, 3\sigma, 4\sigma)$ relative to the control baseline to establish interpretable detection cutoffs. Community detection used the Louvain method (Blondel et al., 2008).

Outlier Detection Procedure. We developed a subject-aware hybrid outlier detection strategy linking multivariate anomalies to specific cohesion signatures. The pipeline combines an Isolation Forest (IF) (F. T. Liu et al., 2008) with a domain-specific Cohesion Composite Score. The IF model used standard hyperparameters (`n_estimators=200`, `contamination=0.01`) to capture general anomalies. The Cohesion Score was computed as

the standardized sum of the four most discriminative features: co-author citation rate, clique strength, reciprocity, and outgoing HHI[1]. We computed 14 behavioural features per author (Table 4). Of these, normalised burst intensity was available for only 7,711 of the 9,431 matched pairs (18% missing) because many authors recorded no burst events in the study window. Statistical tests handle this by dropping incomplete pairs per metric; however, the Isolation Forest requires a complete feature matrix, and zero-imputing 18% of values would inject a spurious signal. The model was therefore trained on the remaining 13 features. All features were standardised (zero mean, unit variance) and multiplied by hand-tuned weights reflecting Random-Forest importance and Wilcoxon effect sizes (e.g., Co-author Rate weighted 4.0); authority metrics were sign-inverted so that lower values indicate greater suspicion. An author was flagged as an outlier only if they satisfied both conditions: an IF anomaly score above the threshold **AND** a Cohesion z-score $> 4\sigma$ (relative to the Control baseline). Sensitivity tests evaluated thresholds from 1σ to 4σ to confirm the robustness of detected syndicates. Outlier labels were merged back into the unified ML table for downstream network analyses (results in Section 4.4).

4 Findings

This section presents the main results of the matched case-control analysis. We first examine overall and field-specific patterns of cohesion and asymmetry, then derive unsupervised behavioral archetypes and identify statistical outliers using a subject-aware Isolation Forest. Together, these analyses reveal that cohesion-rather than sustained one-sided citation flows-is the primary structural distinction between authors publishing in low- and high-influence venues.

Note on metrics. We summarize the main network features used: the *clustering coefficient* (density of local interconnections); *triangles* (closed three-way citation structures); the *k-core*

[1]The Herfindahl-Hirschman Index (HHI) measures concentration. In this context, it quantifies the extent to which an author repeatedly cites the same narrow set of recipients, indicating a lack of source diversity.

number (depth of embedding in dense subnetworks); *betweenness centrality* (how often an author lies on shortest paths, a proxy for brokerage/coordination potential); *velocity* (rate of citation accumulation); *burst* (sudden within-window surges of citations); and *asymmetry* (imbalance in giving vs. receiving citations).

4.1 Overall (All Subjects Combined)

Cohesion dominates. At the aggregate level, authors publishing primarily in low-influence venues display markedly stronger *cohesion*: they cite co-authors **6.7× more frequently** ($p = 9.1 \times 10^{-66}$), exhibit **4.7× higher reciprocity rates** ($p = 2.7 \times 10^{-44}$), and show **3.1× greater outgoing concentration** (tunnel vision, $p \approx 0$). Figure 2 displays the mean paired difference in co-author citation rates across subject fields.

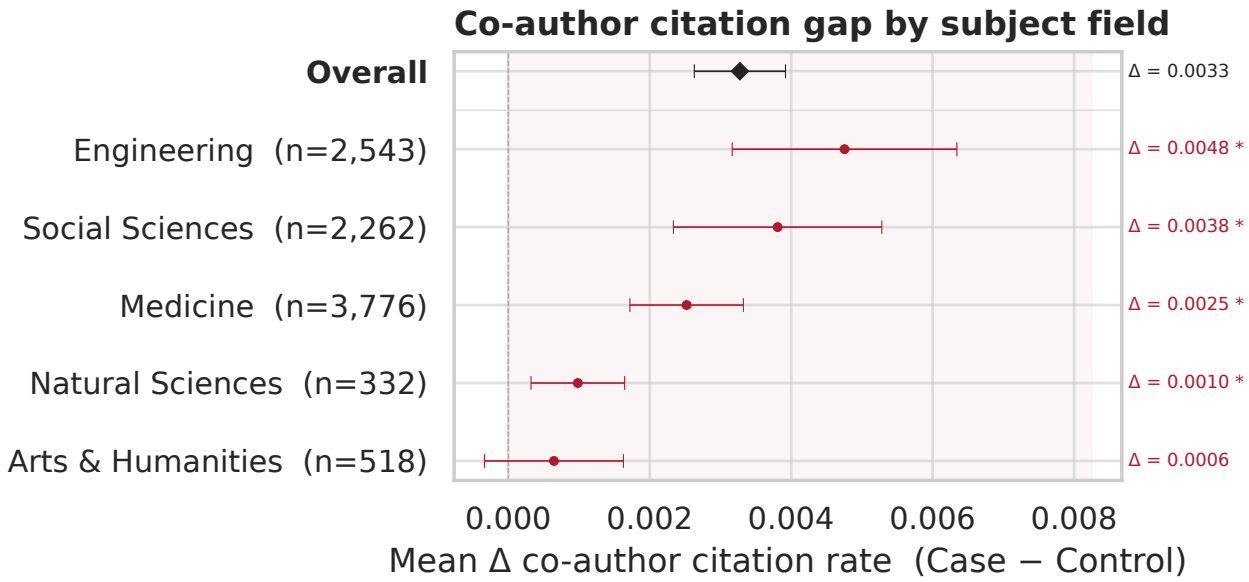

Figure 2: **Co-author citation gap by subject field.** Forest plot of mean paired difference $\Delta = \bar{r}_{\text{Case}} - \bar{r}_{\text{Control}}$ in co-author citation rate, with 95% confidence intervals. Marker • = field estimate; ♦ = overall. Δ labels report rounded effect sizes. The dashed vertical line marks zero (no difference); the pale red shading highlights the Case-higher region.

The cohesion gap is not driven by a few extreme outliers but by a broad rightward shift across all four dimensions (co-author citation rate, self-citation rate, outgoing HHI, and local clustering). Permutation testing (10,000 label shuffles of Case/Control assignments)

independently confirms the effect is far outside the null distribution ($p < 0.001$).

Authority Gap. In contrast to their high internal cohesion, low-influence authors exhibit significantly lower authority metrics. Controls exhibit **3× higher PageRank** than Cases ($p = 3.1 \times 10^{-145}$), and **2.2× higher Citation Entropy**, indicating that while Cases form tight cliques, they lack broad authoritative influence. This divergence is further supported by the random forest feature importances in Figure 3, where cohesion and concentration metrics rank as the most discriminative features for tier classification, while authority metrics like PageRank play a secondary role.

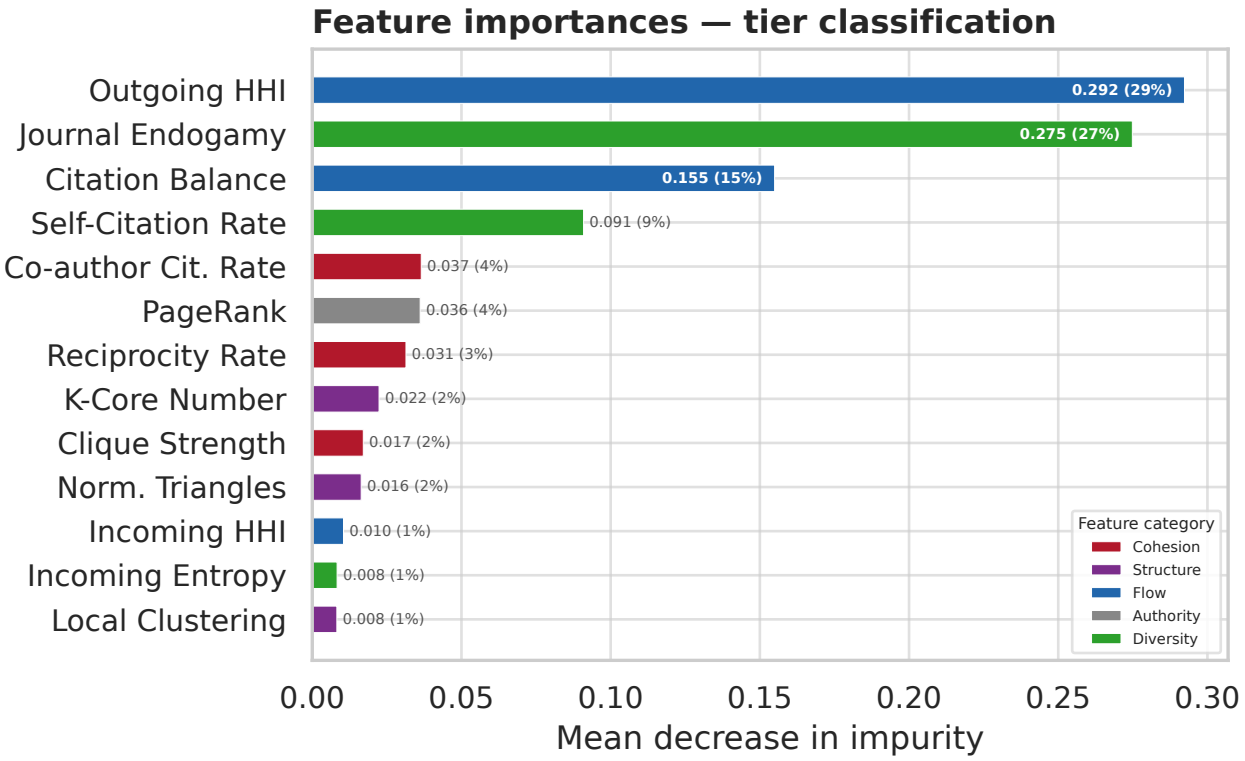

Figure 3: **Random Forest feature importances.** Horizontal bars show mean decrease in impurity for 13 of the 14 behavioural features listed in Table 4, sorted by importance. Bars are coloured by feature category (see legend). Inline labels give raw importance score and share of total. Normalised burst intensity is excluded: it is available for only 82% of pairs, and zero-imputing the missing 18% would inject a spurious signal into the Isolation Forest (Section 3.3).

Net effect. The Bottom-Top gap is explained chiefly by distinct behavioral signatures. These patterns persist under conservative matched-pair comparisons: **12 out of 14** tested

behavioral metrics remain significant after Benjamini-Hochberg FDR correction ($\alpha = 0.05$) (Benjamini & Hochberg, 1995). Crucially, this systemic disparity is not static: cumulative citation trajectories (2020–present) show the gap between Bottom-Tier and Top-Tier authors *widening* over time, indicating an accelerating divergence rather than convergence.

4.2 Subject-Stratified Effects (Subjects 1-5)

Robust cohesion, field-specific accents. Enhanced cohesion is a recurrent feature across subjects, but its expression varies by field (Figure 4). In the health and life sciences, the divergence appears mainly as consistently higher co-author citation and stronger local clustering. In multidisciplinary and physical sciences, dense networking more often coincides with short, episodic activity waves. In the social sciences and humanities, cohesion signals are subtler, but sharper temporal concentrations of citations within short windows are more prominent.

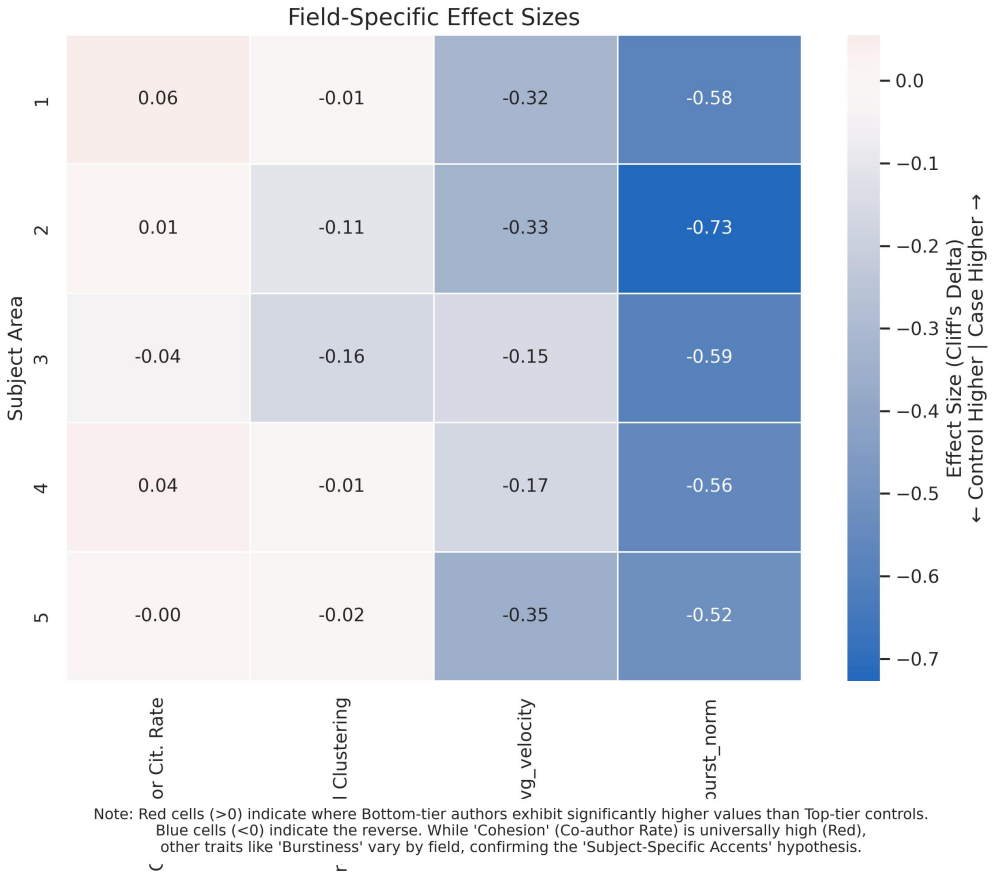

Figure 4: **Subject-Specific Effect Sizes (Cliff's δ).** Each cell reports the effect size for a given metric within a subject area. Positive values (white - pale red) indicate that Cases exhibit higher values than Controls. Negative values (blue) indicate that Controls exhibit higher values than Cases. Co-author Citation Rate shows near-zero or slightly positive values across all subjects, while velocity and burstiness are uniformly negative, with Life Sciences (Subject 2) displaying the largest deficit in burstiness ($\delta = -0.73$).

Muted asymmetry. Across fields, one-sided asymmetry measures remain generally subdued. Where differences emerge, they align more with transient speed-ups (velocity/bursts) than with persistent directional imbalance.

Takeaway. The signature of Bottom-tier authors is greater network cohesion. Whether this manifests as intensified co-author referencing, increased local triadic closure, or episodic waves is field-dependent.

4.3 Behavioral Archetypes (Unsupervised)

K-means clustering on standardized ML features reveals a clear three-way typology: **Asymmetric** (unbalanced citing), **Independent** (typical), and **Solo** (self-citing). Bottom-tier authors (Cases) exhibit multiple higher-cohesion strategies, including Asymmetric and Solo behaviors, while Top-tier Controls converge overwhelmingly on the Independent profile. The Solo self-citing pattern appears almost exclusively among Cases.

This divergence is consistent with clique-like behavior: Case authors follow multiple higher-cohesion strategies, whereas Controls converge on a single low-cohesion, outward-looking profile. Figure 5 illustrates this separation: the Linear Discriminant Analysis (LDA) projection reveals minimal overlap between tiers, with Cases clustering toward the Cohesion-driven extreme and Controls toward the Authority-driven regime.

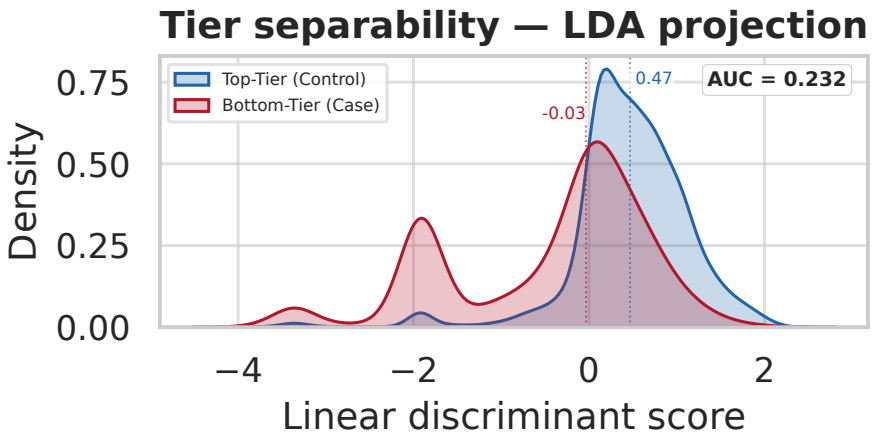

Figure 5: **Tier separability: LDA projection.** Kernel density estimates of the 1-D linear discriminant scores for Bottom-Tier (Case) and Top-Tier (Control) authors. Dotted vertical lines mark within-tier medians. Directional labels indicate the dominant behavioural regime at each tail of the discriminant axis.

Archetype definitions. Three behavioral profiles emerge from the clustering:

- **Asymmetric (Unbalanced)**: high *max_asymmetry* relative to cohesion; concentrated outgoing citation flows.
- **Independent (Typical)**: balanced citing behavior, low cohesion; the dominant profile

among Controls.

- **Solo (Self-citing)**: high *self_citation_rate*; nearly absent among Top-tier authors.

The three-way typology provides interpretable separation: *lower-cohesion* profiles dominate among Controls, while *higher-cohesion and asymmetric* profiles are systematically overrepresented among Cases.

4.4 Outlier Detection Results

Detection Accuracy. Using the hybrid detection strategy at the 4σ threshold, we identified *277 high-confidence outliers*. Of these, **93.5%** are Bottom-tier (Case) authors (*Case purity*), confirming that the pipeline predominantly isolates authors whose structural profiles deviate substantially from the Top-tier norm.

Outlier fingerprint. Relative to non-outlier authors, the detected syndicate members show massive lifts in cohesion signals (Fig. 6): **Clique Strength** $\approx 11\times$, **Co-author Citation Rate** $\approx 6.7\times$, **Reciprocity** $\approx 4.7\times$, and **Tunnel Vision (Outgoing HHI)** $\approx 3.1\times$. Note that the aggregate Case–Control fold-changes reported in Section 4.1 measure the same features across *all* matched pairs, whereas the fingerprint ratios here compare the 277 flagged outliers to the remaining population. These specific deviations constitute the unique *structural fingerprint* of the flagged cohesion outliers.

Table 1: Sensitivity Analysis: Comparison of Detection Methods at 4σ

Method	**Threshold**	**Outliers**	**Case Purity**	**Connected (%)**
Isolation Forest (Baseline)	4σ	269	84.4%	68.8%
Cohesion Composite Only	4σ	599	89.1%	25.7%
Hybrid (IF ∩ Cohesion)	4σ	**277**	**93.5%**	**49.1%**

Note: The Hybrid method combines general multivariate anomalies (IF) with domain-specific cohesion scores, increasing Case Purity by +9.1 percentage points over the baseline.

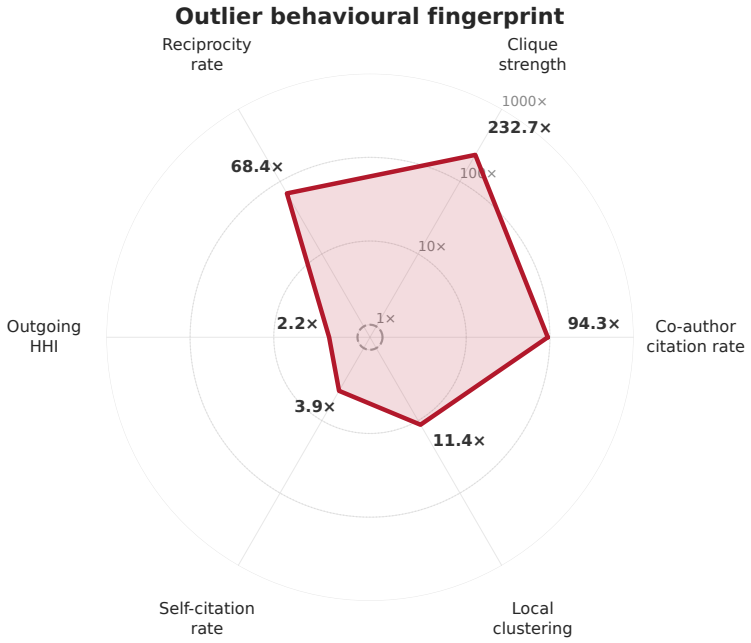

Figure 6: **Outlier behavioural fingerprint.** Radar chart of outlier fold-change ratios (log-scaled radial axis) relative to non-outlier authors across six citation-behaviour metrics. Values on each spoke give the outlier-to-normal mean ratio; concentric rings mark 1×, 10×, and 100× baselines. The dashed inner ring corresponds to 1× (no difference).

Table 1 demonstrates the superiority of the hybrid approach. While Isolation Forest alone achieves 84.4% purity, augmenting it with the cohesion filter improves purity to 93.5%. This indicates that approximately 9% of statistical outliers are legitimate prolific collaborators, who are successfully filtered out by the cohesion requirement.

4.5 Network Forensics (Largest Outlier Component)

The largest outlier syndicate ($n = 23$) exhibits a distinct hub-and-spoke topology rather than a dense, uniform mesh. Within this group, the flow imbalance analysis reveals a clear hierarchy: a single **"Primary Beneficiary"** occupies the central position, receiving directed citation flows from peripheral **"Sycophants"** who define themselves by giving significantly more citations than they receive.

Temporal activity supports the orchestration hypothesis, showing a sudden-burst pattern with high internal citation intensity. As shown in the timeline analysis, citation accumulation is not organic but driven by coordinated waves, particularly evident in 2021 and 2024.

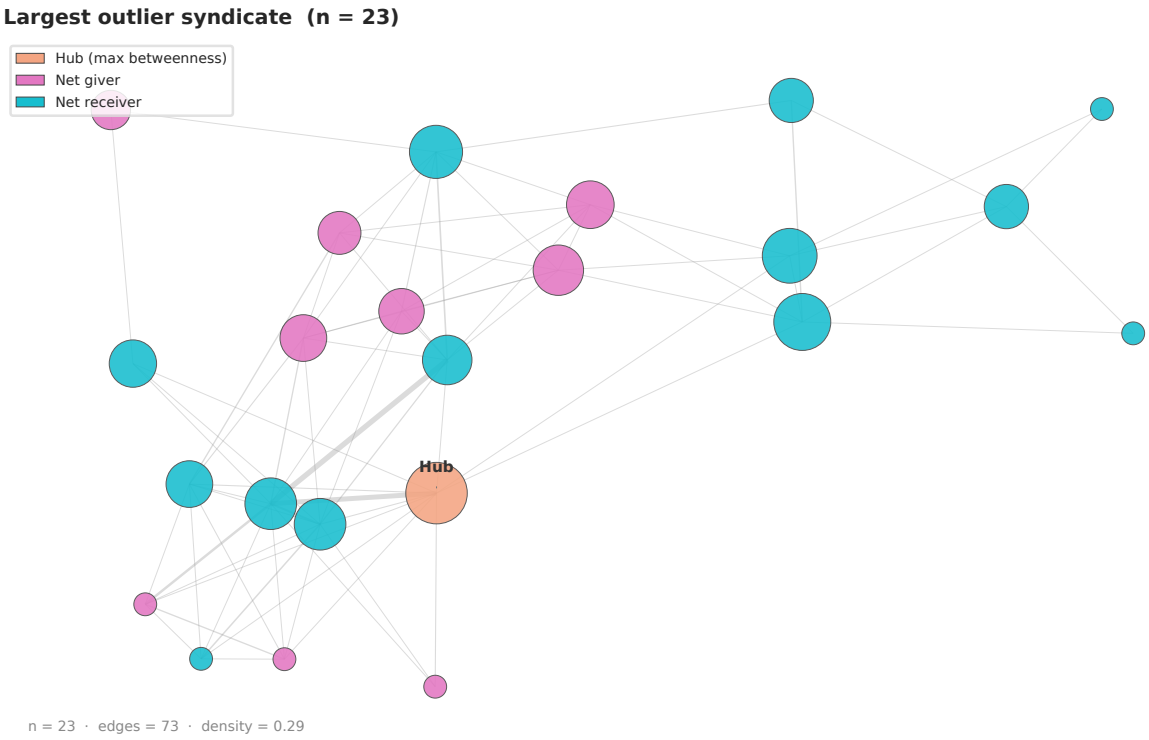

Figure 7: **Largest outlier citation syndicate ($n = 23$).** Force-directed layout of the internal citation network. Node size is proportional to betweenness centrality; salmon = hub (maximum betweenness); red = syndicate member. Edge width scales with aggregate citation weight. The inset box summarises graph statistics.

The hub-and-spoke topology and coordinated temporal bursts reveal a pervasive "Sycophant-Beneficiary" flow imbalance that distinguishes Cases from Controls.

4.6 Discussion

Segregation and the "Two Worlds" Phenomenon. The distinction between low- and high-influence venues extends beyond simple density; it reflects extreme *segregation*. The high assortativity coefficient ($r = 0.71$) and modularity ($Q = 0.97$) confirm a **"Two Worlds" hypothesis**, where Bottom-tier authors operate in closed citation economies with minimal external engagement. Flagged outliers concentrate at the high-asymmetry, high-burst corner of the anomaly landscape, a region essentially unpopulated by Controls. In this sense, citation cliques are not merely organic clusters but isolated silos disconnected from the broader scientific discourse.

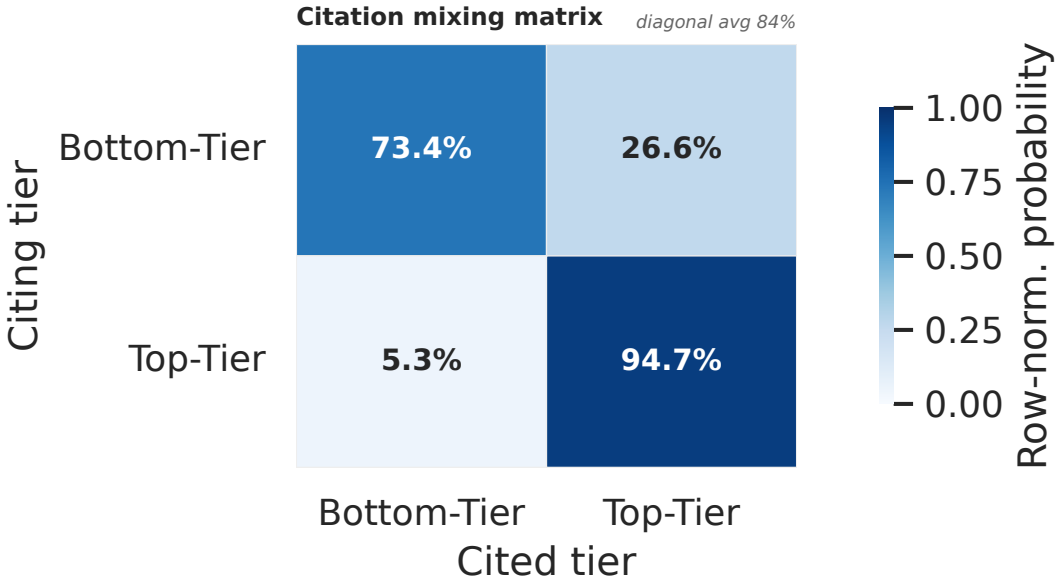

Figure 8: **Citation mixing matrix ($r = 0.71$, $Q = 0.97$; strong homophily).** Row-normalised proportion of citations directed within and across tiers. Cell values show the conditional probability that an author in the row tier cites an author in the column tier. The Brewer RdBu colormap is centred at 0.5. Below: diagonal average (84.0%) summarises overall within-tier preference.

4.7 Cross-referencing Suspicious Authors

To move from aggregate outlier statistics to actionable intelligence, we cross-reference the 277 hybrid outliers against real-world author identities and publication records. Of these, 189 (68.2%) had publicly resolvable ORCID profiles and are analysed below; the remaining

88 lacked public ORCID records or returned incomplete metadata. Each ORCID identifier is resolved to a name via the ORCID public API,[2] and publication portfolios are reconstructed from `impact.db` by joining `work_authors` to `works` and resolving journal titles through the `journal_names` catalogue.

Suspiciousness scoring. We define a composite *suspiciousness score* as the weighted sum of population-normalised z-scores across six behavioural features:

$$S_i = \sum_{k=1}^{6} w_k \cdot z_{ik}, \qquad z_{ik} = \frac{x_{ik} - \bar{x}_k}{\sigma_k}, \tag{1}$$

where x_{ik} is the raw value of feature k for author i, $\bar{x}_k$ and σ_k are the population mean and standard deviation, and the weights w_k are chosen to reflect both Random-Forest feature importance (Figure 3) and Wilcoxon effect sizes: co-author citation rate ($w = 4.0$), clique strength (3.5), reciprocity rate (3.5), outgoing HHI (3.0), self-citation rate (2.0), and journal endogamy rate (2.0).

Authors are ranked by S_i in descending order. An individual feature is flagged when its z-score exceeds 3σ or 5σ.

Key Findings. Of the 189 outliers, 162 (85.7%) belong to the Case (bottom-tier) group, confirming the precision of the Isolation Forest detector. Outliers concentrate in Health Sciences ($n = 77$) and Physical Sciences ($n = 71$), followed by Social Sciences ($n = 46$) and Life Sciences ($n = 3$); per-subject counts sum to 197 because eight authors are flagged in two subjects each. A total of 131 outliers (69.3%) belong to at least one syndicate (connected component among mutual outlier citation links), with the largest syndicate comprising 23 members (see Figure 7).

Table 2 presents the ten highest-scoring authors. All ten exhibit co-author citation rates exceeding 5σ; seven also show reciprocity rates beyond 5σ. The two top-ranked authors –both in agricultural engineering– share an identical suspiciousness score ($S = 658.6$) and belong to the same 23-member syndicate, consistent with coordinated behaviour. The presence of

[2] https://pub.orcid.org

Table 2: Top-10 most suspicious authors ranked by composite suspiciousness score S (Eq. 1).

Rank	ORCID	Name	Subject	S	Works	Primary red flags
1	0000-0001-5794-5580	O. Kochuk-Yashchenko	Eng.	658.6	12	co-auth. cit., clique str., recip. ($> 5\sigma$)
2	0000-0002-1998-6290	D. Kucher	Eng.	658.6	13	co-auth. cit., clique str., recip. ($> 5\sigma$)
3	0000-0001-9044-3400	E. Ivanova	Med.	361.7	4	co-auth. cit., recip., out. HHI ($> 5\sigma$)
4	0000-0001-5541-5180	L. A. dos Santos Franz	Soc.	207.6	14	co-auth. cit. ($> 5\sigma$); self-cit. ($> 3\sigma$)
5	0000-0002-0836-9510	J. Ryu	Eng.	204.9	4	co-auth. cit., clique str., recip. ($> 5\sigma$)
6	0000-0002-5354-3020	S. H. Lee	Eng.	203.4	5	co-auth. cit., clique str., recip. ($> 5\sigma$)
7	0000-0002-5962-4470	M. L. Grigio	Eng.	203.0	5	co-auth. cit., recip., out. HHI ($> 5\sigma$)
8	0000-0002-3497-8870	K. Nam	Eng.	184.7	4	co-auth. cit., recip. ($> 5\sigma$); endog. ($> 5\sigma$)
9	0000-0001-6877-9530	D. R. Amirov	Eng.	180.4	5	co-auth. cit., recip. ($> 5\sigma$); self-cit. ($> 3\sigma$)
10	0000-0003-1664-6810	D. Morozova	Eng.	180.4	4	co-auth. cit., recip. ($> 5\sigma$); self-cit. ($> 3\sigma$)

Note. Suspiciousness score is a weighted z-score composite of six behavioural features. Red flags indicate features exceeding 3σ or 5σ of the population mean. These patterns are *statistically anomalous*; they do not constitute proof of misconduct. Names are initialised for brevity; full ORCIDs are verifiable via `https://orcid.org/[ORCID]`.

authors writing in Korean, Russian, Ukrainian, and Portuguese underscores the international scope of the phenomenon.

Table 3 provides a publication-level audit of the same ten authors. Typical portfolios are small (4-14 works, median 5), concentrated in a single regional journal, and characterised by near-zero outgoing citations in the global network yet elevated reciprocal citation links, a pattern consistent with closed-loop mutual citation within a small community of co-authors.

Implications for editorial practice. The concentration of suspicious behaviour in small, regional journals many of which do not appear in major indexing databases, suggests that coordinated citation inflation is disproportionately prevalent in venues with limited editorial oversight. The publication-audit data show that the highest-scoring authors maintain very small portfolios (median 5 works), yet achieve anomalous citation metrics through closed-loop mutual referencing. This pattern is consistent with prior work on "citation stacking" (Heneberg, 2016) and suggests that journal-level surveillance of incoming self-citation rates from small author cliques could serve as an effective early-warning signal.

Table 3: Publication audit profiles for top-10 suspicious authors.

Rank	ORCID	Works	Years	Out	In	Recip.	Primary journal
1	0000-0001-5794-5580	12	2020–2024	0	1	1	Bull. Sumy Nat. Agrar. Univ.
2	0000-0002-1998-6290	13	2020–2024	0	1	1	Anim. Breed. Genet.
3	0000-0001-9044-3400	4	2020–2023	1	1	1	Laser Medicine
4	0000-0001-5541-5180	14	2020–2024	0	0	0	Exacta
5	0000-0002-0836-9510	4	2020–2022	0	0	1	Korean J. Soil Sci. Fert.
6	0000-0002-5354-3020	5	2020–2022	0	0	1	Korean J. Soil Sci. Fert.
7	0000-0002-5962-4470	5	2020–2022	0	0	1	Acta Sci. Agronomy
8	0000-0002-3497-8870	4	2021–2023	0	0	1	Korean J. Plant Taxon.
9	0000-0001-6877-9530	5	2021–2023	0	0	1	Sci. Notes Kazan Vet. Acad.
10	0000-0003-1664-6810	4	2022–2023	0	0	1	Int. J. Vet. Med.

Note. Out = total outgoing citations in the directed citation network; In = total incoming; Recip. = reciprocal citation pairs. "Primary journal" is the journal in which the author published the most works within the 2020–2024 study window.

Internal Hierarchy: Beneficiaries and Sycophants. While broad asymmetry metrics are generally subdued, the internal structure of detected syndicates reveals a distinct **flow imbalance**. As seen in the largest syndicate ($n = 23$), a hub-and-spoke topology emerges where peripheral members ("Sycophants") act as net givers to central "Beneficiaries". This suggests that the observed cohesion serves a functional purpose: the coordinated funneling of credit to central figures, rather than reciprocal exchange among equals.

Relation to prior literature. These findings extend the concept of *citation orchestration* (Evdaimon, Ioannidis, et al., 2024), showing that orchestration manifests as persistent reciprocity and dense closure. The limited cross-tier mixing supports concerns about *home bias* (Qiu et al., 2025) and institutional siloing. However, the specificity of our *hybrid detection* (93.5% purity) suggests that these patterns in low-influence venues are structural anomalies distinct from legitimate topic specialization.

Implications for evaluation. The results highlight that aggregate counts hide structural manipulation. Evaluation systems must incorporate **cohesion-aware metrics** –specifically *co-author citation rates* and *flow imbalance*– to distinguish organic collaboration from dense internal referencing. The success of the hybrid pipeline suggests that combining multivariate anomaly detection (Isolation Forest) with targeted cohesion signals can effectively flag insular publication ecosystems.

Broader implications. Overall, the dominance of segregation and flow imbalance suggests that low-influence venues foster **closed economies** of citation. Understanding these patterns is essential for designing fair bibliometric indicators that reward genuine impact rather than inward-looking visibility loops.

5 Limitations and Future Work

This study relies on Crossref metadata (2020–2024), which vary in completeness. While the hybrid filter significantly reduces false positives, the distinction between a "cartel" and a hyper-specialized research group remains a probabilistic assessment. The LLM-based journal classification (Appendix D) was not validated against an independent ground truth; misclassification of some journals could shift the boundary between Bottom- and Top-tier authors. Moreover, citation links are analyzed structurally, without text-level context to infer the motivation of citing. Finally, the network analysis is conducted at a structural level and does not expose the identities of the individual authors involved; flagged nodes are treated as anonymous structural roles (e.g., "Beneficiary" or "Sycophant") rather than named individuals.

Ethical considerations. Where individual author profiles are discussed (Tables 2 and 3), the underlying data consist exclusively of publicly available Crossref metadata and self-declared ORCID identifiers. Identifying anomalous citation patterns in publicly funded research serves a legitimate accountability function, consistent with established practice in the retraction and research-integrity literature (Besançon, Cabanac, et al., 2024; Biagioli & Lippman, 2020). Author names are initialised and no institutional affiliations are disclosed; readers can verify the underlying records through the cited ORCID links.

Future work could combine the structural fingerprint with author disambiguation and profile-level inspection to move from structural suspicion to targeted auditing. Extending the temporal coverage and triangulating with independent data sources (e.g., OpenAlex) could

clarify whether the observed "Two Worlds" segregation is widening over time. Experimental audits in selected venues could also test whether editorial interventions can dismantle the "Echo Chambers" identified in this analysis.

6 Summary and Conclusions

We developed a subject-aware hybrid pipeline that integrates LLM-based journal classification, per-subject Eigenfactor scores, and a composite outlier detection strategy. This design allowed us to isolate specific behavioral signatures once productivity (by subject-specific h_5) is held constant.

The evidence points to a highly structured phenomenon. In the aggregate Case–Control comparison, authors publishing predominantly in low-influence journals cite co-authors **6.7× more frequently**; among the 277 flagged outliers the lift is even starker, with **11× greater clique strength** relative to non-outliers. Rather than random noise, outlier syndicates display a **hub-and-spoke topology**, where peripheral "Sycophants" funnel citations to central "Beneficiaries" through coordinated bursts. The flow imbalance analysis confirms that Case authors systematically give more citations than they receive. Cumulative citation trajectories over the 2020–2024 window suggest that the gap between tiers may be widening, although a longer observation period is needed to confirm this trend.

Controlling for subject and productivity, the hybrid detection strategy identifies structurally anomalous citation groups with 93.5% Case purity (proportion of flagged authors belonging to the Bottom tier). These groups are characterized not by simple asymmetry, but by a specific cohesion fingerprint –extreme co-author citation rates, dense local clustering, and tunnel vision. Consequently, low-influence venues do not merely host lower-quality work; they foster **segregated, inward-looking citation economies** whose structural properties are associated with distortion of bibliometric indicators.

7 Author Contributions

Conceptualization: P.-A.S., D.S.; Methodology: P.-A.S.; Software: P.-A.S., G.A.; Data curation: P.-A.S., G.A.; Validation: P.-A.S., G.A., D.S.; Formal analysis: P.-A.S., G.A.; Visualization: P.-A.S.; Writing – original draft: G.A.; Writing – review & editing: P.-A.S., G.A., D.S.; Supervision: D.S.; Project administration: D.S.

8 Competing Interests

The authors declare no conflict of interest.

9 Funding Information

This research did not receive a specific grant from any funding agency in the public, commercial or non-profit sectors.

10 Data and Code Availability

All essential data and code for this study are openly available at Zenodo under a CC BY license (Spanakis et al., 2026). The analysis is implemented in a single, self-contained script (`citation_analysis.py`) that reproduces all results in one invocation: it loads the pre-built `rolap.db` and `impact.db` databases, runs the statistical comparisons (Section 4.1), generates all eight publication figures, and produces the author suspiciousness tables (Section 4.7).

References

Abalkina, A., Aquarius, R., Bik, E. M., Bimler, D., Bishop, D., Byrne, J., Cabanac, G., Day, A., Labbé, C., & Wise, N. (2025). Stamp out paper mills. *Nature*, *637*, 1047–1049. https://doi.org/10.1038/d41586-025-00212-1

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, *57*(1), 289–300.

Bergstrom, C. T., West, J. D., & Wiseman, M. A. (2008). The eigenfactor metrics: A network approach to assessing scholarly journals. *Journal of Neuroscience*, *28*(45), 11433–11434. https://doi.org/10.1523/JNEUROSCI.0003-08.2008

Besançon, L., Cabanac, G., et al. (2024). When scientific citations go rogue: Uncovering sneaked references. *Journal of the Association for Information Science and Technology*. https://doi.org/10.1002/asi.24896

Biagioli, M., & Lippman, A. (Eds.). (2020). *Gaming the metrics: Misconduct and manipulation in academic research*. MIT Press. https://doi.org/10.7551/mitpress/11087.001.0001

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, (10), P10008.

Cliff, N. (1993). *Dominance statistics: Ordinal analyses to answer ordinal questions*. Psychological Assessment Resources.

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd). Routledge.

Efron, B., & Tibshirani, R. J. (1993). *An introduction to the bootstrap*. Chapman & Hall.

Evdaimon, I., Ioannidis, J. P. A., et al. (2024). Metrics to detect small-scale and large-scale citation orchestration. *arXiv Cornell University*. https://doi.org/10.48550/arXiv.2406.19219

Heneberg, P. (2016). From excessive journal self-cites to citation stacking: Analysis of journal self-citation kinetics in search for journals, which boost their scientometric indicators. *PLoS ONE*, *11*(4), e0153730. https://doi.org/10.1371/journal.pone.0153730

Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008). Isolation forest. *Proceedings of the 8th IEEE International Conference on Data Mining*, 413–422.

Liu, J., Bai, X., Wang, M., Tuarob, S., & Xia, F. (2024). Anomalous citations detection in academic networks. *Artificial Intelligence Review*, *57*(4), 103. https://doi.org/10.1007/s10462-023-10655-5

LLM Guidelines Consortium. (2024). *Llm guidelines: Transparency in llm-assisted research*. Retrieved September 17, 2025, from https://llm-guidelines.org/

Naddaf, M. (2025). How to spot fake scientists and stop them from publishing papers [News Feature, 22 October 2025]. *Nature*, *646*, 792–794. https://doi.org/10.1038/d41586-025-03341-9

Qiu, S., Steinwender, C., & Azoulay, P. (2025). *Paper tiger? chinese science and home bias in citations* (NBER Working Paper No. 32468) (Revised from May 2024 version). National Bureau of Economic Research. https://doi.org/10.3386/w32468

Spanakis, P.-A., Alexandrou, G., & Spinellis, D. (2026). *Replication data for: Citation cliques in questionable journals* (Version v1). Zenodo. https://doi.org/10.5281/zenodo.xxxxxxx

Spinellis, D. (2023). Open reproducible scientometric research with alexandria3k. *PLoS ONE*, *18*(11), e0294946. https://doi.org/10.1371/journal.pone.0294946

Spinellis, D. (2024). Unit tests for sql. *IEEE Software*, *41*(1), 31–34. https://doi.org/10.1109/MS.2023.3328788

Wilcoxon, F. (1945). Individual comparisons by ranking methods. *Biometrics Bulletin*, *1*(6), 80–83.

A Feature Definitions

Table 4 provides formal definitions for all behavioural metrics used in the analysis. In the formulas below, w_{ij} denotes the fractional citation weight from author i to author j (see Appendix B), and sums run over non-self edges unless stated otherwise.

Table 4: **Behavioural feature definitions.** All metrics are computed per author over the 2020–2024 citation network. N_{out} = number of distinct cited authors; N_{in} = number of distinct citing authors; $p_j = w_{ij}/\sum_k w_{ik}$ (outgoing share); $q_j = w_{ji}/\sum_k w_{ki}$ (incoming share).

Feature	**Formula**	**Interpretation**
Self-citation rate	$\frac{\sum w_{ii}}{\sum_j w_{ij}}$	Share of outgoing citation weight directed at the author's own works.
Co-author citation rate	$\frac{\sum_{j \in C_i} w_{ij}}{\sum_{j \neq i} w_{ij}}$	Among non-self citations, proportion citing former co-authors (C_i).
Citation balance	$\frac{w_{\text{out}} - w_{\text{in}}}{w_{\text{out}} + w_{\text{in}} + \epsilon}$	Net giver/receiver bias; positive $\Rightarrow$ net giver. Bounded $(-1, +1)$, comparable across activity levels.
PageRank	weighted PR on G	Global influence score computed via weighted PageRank on the directed citation graph.
Journal endogamy	$\frac{\lvert\{\text{refs with same ISSN}\}\rvert}{\lvert\{\text{all refs}\}\rvert}$	Proportion of references citing papers in the same journal.
Citation entropy	$H = -\sum_j q_j \ln q_j$	Shannon diversity of incoming citation sources. Low $\Rightarrow$ concentrated.
Citation HHI (in)	$\sum_j q_j^2$	Herfindahl–Hirschman concentration of incoming citations.
Reciprocity rate	$\frac{\lvert\{j : w_{ij} > 0 \wedge w_{ji} > 0\}\rvert}{\lvert\{j : w_{ij} > 0\}\rvert}$	Share of cited peers who also cite back.
Outgoing HHI	$\sum_j p_j^2$	Concentration of outgoing citations ("tunnel vision").
Clustering coeff.	$\frac{2\,\lvert\{(j,k) \colon j,k \in \mathcal{N}_i,\ (j,k) \in E\}\rvert}{d_i(d_i - 1)}$	Fraction of an author's neighbours that are themselves connected. Computed on the undirected projection of the internal citation graph.
Triangles (norm.)	$\frac{\Delta_i}{\sum_j w_{ij} + \sum_j w_{ji} + 1}$	Triangle count normalised by total citation activity.
k-core number	$\max\{k : i \in H_k\}$	Largest k such that author i belongs to the k-core of the undirected citation graph.

Continued on next page

Table 4 continued

Feature	Formula	Interpretation
Clique strength	clustering × co-author citation rate	Joint cohesion signal: dense neighbourhood *and* preferential co-author citing.
Norm. burst intensity†	$\frac{\max_j w_{ij}}{\text{in_strength} + 1}$	Largest single-peer citation weight normalised by total incoming volume. Available for 7,711 of 9,431 pairs; excluded from the Random Forest owing to 18% missing data (see Section 3.3).

B Matching Procedure and Detection Parameters

B.1 Author Classification

Authors were classified based on their publication portfolio:

- **Bottom Tier**: $\geq 70\%$ of papers published in journals below the 25th subject-specific Eigenfactor percentile, with a minimum of 3 papers.
- **Top Tier**: $\geq 70\%$ of papers published in journals above the 75th subject-specific Eigenfactor percentile, with a minimum of 3 papers.
- **Other**: authors not meeting either threshold.

B.2 Matched-Pair Construction

Case–Control pairs were formed via greedy one-pass matching within each subject area. For each Bottom-Tier author the algorithm selected the closest available Top-Tier author by subject-specific h_5 index. Productivity was bucketed by integer division ($b = \lfloor h_5/3 \rfloor$); candidates with $|b_{\text{case}} - b_{\text{control}}| \leq 1$ (i.e., ± 3 h_5 points) were eligible. Among eligible candidates the one minimising $|h_{5,\text{case}} - h_{5,\text{control}}|$ was selected. Each (ORCID, subject) pair was used

at most once (matching without replacement). This procedure yielded 9,431 matched pairs across five subject areas (Table 6).

B.3 Fractional Citation Weights

Each citation edge between a citing work with n_c authors and a cited work with n_a authors received weight

$$w = \frac{1}{n_c \times n_a}\,.$$

Self-loops ($i = j$) were retained and tracked via a Boolean flag for the self-citation rate metric; all other metrics were computed on non-self edges. The temporal window covers works published between 2020 and 2024 inclusive.

B.4 Outlier Detection

Table 5 summarises the Isolation Forest hyper-parameters. The model was trained per-subject on 13 of the 14 scaled features listed in Table 4 (excluding normalised burst intensity; see Section 3.3).

Table 5: **Isolation Forest hyper-parameters.**

Parameter	**Value**
`n_estimators`	200
`contamination`	0.01
`max_samples`	`auto` ($\min(256, n)$)
`max_features`	1.0 (all features)
`bootstrap`	False
`random_state`	42

Cohesion composite score. A secondary filter computes a weighted z-score relative to the Control population:

$$S_i = \sum_{k=1}^{6} w_k \cdot z_{ik}, \qquad z_{ik} = \frac{x_{ik} - \bar{x}_k}{\sigma_k},$$

with weights: co-author citation rate (4.0), clique strength (3.5), reciprocity rate (3.5), outgoing HHI (3.0), self-citation rate (2.0), journal endogamy (2.0). Standardisation uses the population mean and standard deviation. An author was flagged only if both the Isolation Forest anomaly score exceeded its per-subject threshold *and* $S_i > 4\sigma$.

C Data Derivation and Sampling

The analytical dataset was derived through a multi-stage pipeline. This appendix documents the full data flow and the filtering decisions at each stage.

C.1 Stage 1: Raw Data Ingestion

Crossref metadata were ingested via *alexandria3k* (Spinellis, 2023), retaining works published between 2020 and 2024. The following fields were extracted: DOI, publication year, page information, print and electronic ISSN, reference lists (DOI-level), and author ORCID identifiers.

C.2 Stage 2: Journal-Level Metrics

Works were linked to journals through normalised ISSNs. A citability filter removed non-article items (single-page entries such as editorials and news pieces). For each journal, two-year and five-year citation means, the h_5 index, network centrality (Eigenfactor), prestige rank (SJR-like), mean article score, and context-normalised impact were computed. Journal citation networks were constructed from references in the reference year (2024) to publications

in the preceding two-, three-, or five-year windows, depending on the metric.

C.3 Stage 3: Journal Subject Classification

Each journal was assigned to one of five broad subjects using an LLM-based classifier (see Appendix D). The final mapping covers approximately 15,000 ISSNs. Within each subject, the 25th and 75th Eigenfactor percentiles were computed to define *questionable* and *high-influence* thresholds, respectively.

C.4 Stage 4: Author Population and Sampling

To keep computation tractable, the author population was sampled by retaining only ORCIDs whose last digit is `0` (approximately 10% of all ORCID-bearing authors in the Crossref snapshot). This deterministic filter preserves the distribution of author characteristics because the ORCID check digit is algorithmically assigned and uncorrelated with research behaviour.

C.5 Stage 5: Author Classification and Matching

Within the sampled population, authors were classified per subject as:

- **Bottom Tier**: $\geq 70\%$ of papers in journals below the 25th Eigenfactor percentile, with ≥ 3 papers.
- **Top Tier**: $\geq 70\%$ of papers in journals above the 75th Eigenfactor percentile, with ≥ 3 papers.

Authors not meeting either criterion were excluded. A subject-specific h_5 index was computed for each remaining author. Bottom-Tier authors were then matched to Top-Tier authors via greedy one-pass matching within subject, as described in Appendix B.

C.6 Stage 6: Final Dataset

The matching procedure yielded 9,431 case-control pairs across five subjects (Table 6). For the maximum citation burst metric, 7,711 pairs were available due to data sparsity. Directed author-to-author citation networks were constructed for matched authors using fractional edge weights (Appendix B).

Table 6: **Matched pairs by subject area.**

Subject	**Pairs**
Health Sciences	3,776
Life Sciences	2,543
Physical Sciences	2,262
Social Sciences & Humanities	518
Multidisciplinary	332
Total	**9,431**

D LLM-Based Journal Classification

Journal subject assignment used OpenAI `gpt-4.1` with temperature 0 and deterministic seeding. Each journal was classified into one of five broad categories: (1) Health Sciences, (2) Life Sciences, (3) Multidisciplinary, (4) Physical Sciences, (5) Social Sciences & Humanities.

The prompt supplied the journal title and, where available, a representative abstract, together with few-shot exemplars for each category. Decision rules prioritised abstract content over titles and journal names to reduce ambiguity. Predictions were cached deterministically to a JSON file; the final mapping covers approximately 15,000 ISSNs stored in the `issn_subjects` table of the study database. No independent ground-truth validation was performed; this is acknowledged as a limitation.